A FAMILY OF CLOSED SUBSETS OF A QUASI-COMPACT SPACE WITH THE FINITE INTERSECTION PROPERTY HAS NONEMPTY INTERSECTION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of quasi-compactness through the finite-subcover property of open coverings, and the complementation correspondence between open and closed subsets. Set up in the author's section on quasi-compact spaces and maps, immediately before the counted statement. Supplies the reformulation of the statement that the proof argues by contraposition.

Lemma 0.1. *Let X be a quasi-compact topological space. If $\{Z_\alpha\}_{\alpha\in A}$ is a collection of closed subsets such that the intersection of each finite subcollection is nonempty, then $\bigcap_{\alpha\in A} Z_\alpha$ is nonempty.*

Proof. We suppose that $\bigcap_{\alpha\in A} Z_\alpha = \emptyset$. So we have $\bigcup_{\alpha\in A}(X \setminus Z_\alpha) = X$ by complementation. As the subsets Z_α are closed, $\bigcup_{\alpha\in A}(X \setminus Z_\alpha)$ is an open covering of the quasi-compact space X. Thus there exists a finite subset $J \subset A$ such that $X = \bigcup_{\alpha\in J}(X \setminus Z_\alpha)$. The complementary is then empty, which means that $\bigcap_{\alpha\in J} Z_\alpha = \emptyset$. It proves there exists a finite subcollection of $\{Z_\alpha\}_{\alpha\in J}$ verifying $\bigcap_{\alpha\in J} Z_\alpha = \emptyset$, which concludes by contraposition. □

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